

## A Plumber's Choice

Answers

You need to organise a plumber after a burst pipe has flooded your bedroom. You have been given three quotes for replacing the broken pipes.

### Choice 1 – Pete's Plumbing

Pete charges \$60 per hour plus \$150 flat fee (start up fee)

### Choice 2 - Wendy's Water Workers

Wendy charges \$80 per hour plus \$100 flat fee

### Choice 3 – Lou's Leaks

Lou charges \$100 per hour and but does not charge an initial start up fee

1. (a) Write down a linear equation to represent what each plumber would cost in \$ (c) given the number of hours (n) they work for.
- (b) Complete the table of values for each plumber

#### Pete's Plumbing

Linear Equation:  $c = 60n + 150$

|                     |     |     |     |     |     |     |     |
|---------------------|-----|-----|-----|-----|-----|-----|-----|
| N (number of hours) | 0   | 1   | 2   | 3   | 4   | 5   | 6   |
| C (cost in \$)      | 150 | 210 | 270 | 330 | 390 | 450 | 510 |

To help you, use this equation template

$$\text{Cost} = 60 \times \text{hours} + 150$$

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$$\text{Cost} = \underline{\quad}$$

#### Wendy's Water Workers

Linear Equation:  $c = 80n + 100$

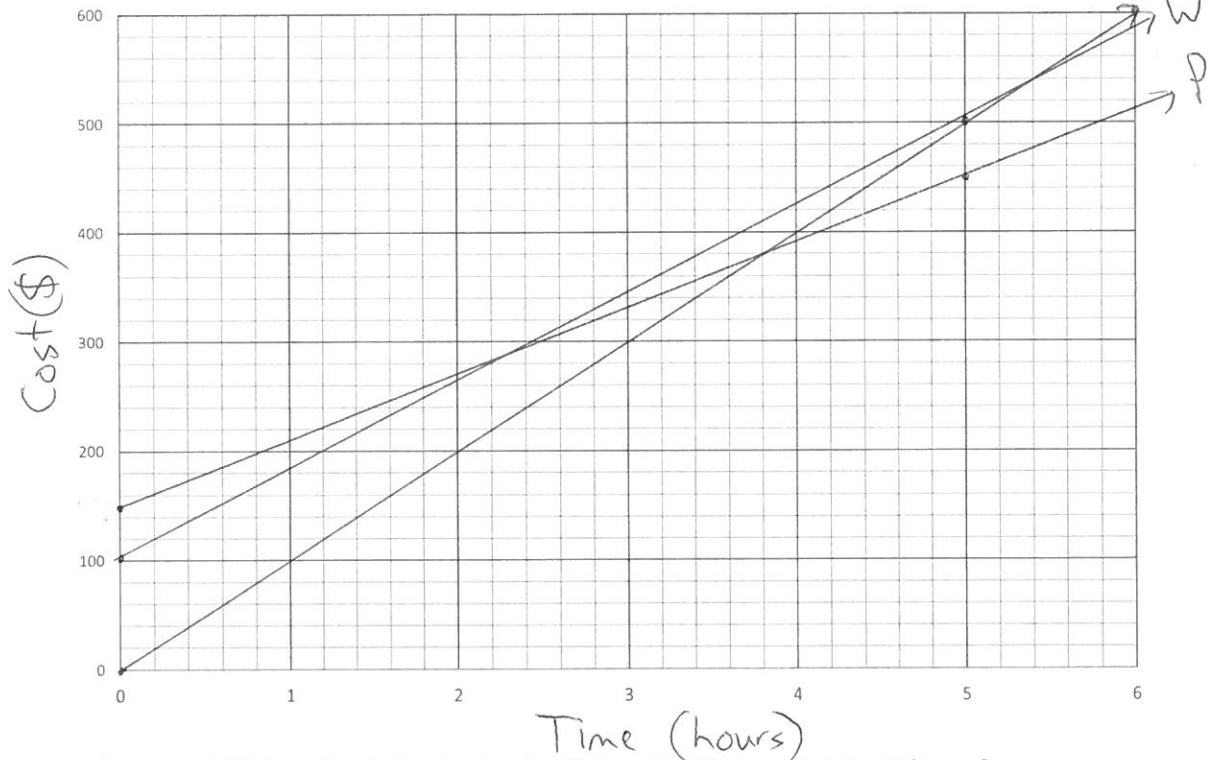
|                     |     |     |     |     |     |     |     |
|---------------------|-----|-----|-----|-----|-----|-----|-----|
| N (number of hours) | 0   | 1   | 2   | 3   | 4   | 5   | 6   |
| C (cost in \$)      | 100 | 180 | 260 | 340 | 420 | 500 | 580 |

#### Lou's Leaks

Linear Equation:  $c = 100n + 0$

|                     |   |     |     |     |     |     |     |
|---------------------|---|-----|-----|-----|-----|-----|-----|
| N (number of hours) | 0 | 1   | 2   | 3   | 4   | 5   | 6   |
| C (cost in \$)      | 0 | 100 | 200 | 300 | 400 | 500 | 600 |

2. Plot these three options on the graph paper provided and label each option.



3. Which option is the best value if the plumbing work takes 2 hours?

Lou's Leaks - \$200

4. Which option is the best value if the plumbing work takes 5 hours?

Pete's Plumbing - \$450

5. An estimate has suggested that the work should take 4 and a half hours. Using your graph, find out what the cost would then be for each of the three options.

P - \$420  
W - \$450  
L - \$450

6. If you have a budget of \$300 which plumber could you hire for the longest, and how long could you hire them for?

Lou's - \$300

7. How does your answer change if your budget increases to \$700?

Pete's  $\$750 = 60n + 150$   
 $\$600 = 60n$   
 $n = 10$

8. Wendy's Water Workers is struggling for business. Why is this?  
 Never the cheapest